

How do different project delivery methods shape communication networks among owners, designers, and contractors in construction projects?

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On my honor as a University Student, I have neither given nor received unauthorized aid on this assignment as defined by the Honor Guidelines for Thesis-Related Assignments

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INTRODUCTION

Miscommunication is one of the most common causes of delays and increased costs in construction projects because of the need for precise technical coordination among multiple teams. Beyond simple information exchange, these challenges are rooted in how relationships between stakeholders are structured, influencing trust, collaboration, and the willingness to communicate openly (Yang et al., 2023). The main stakeholders involved are the owners, the architects, and the general contractor. Their roles of directing, designing, or executing the work remain consistent across projects, but responsibility for key decisions varies depending on the contractual relationships defined by the project delivery method chosen for the project. Project delivery methods establish the hierarchy of a project and determine each stakeholder's level of involvement and authority, creating the structure through which communication, trust, and collaboration occur (Engebø et al., 2020).

Construction projects rely on exchanging large amounts of highly detailed technical information through drawings, specifications, and submittals formal documents contractors provide to confirm materials and methods meet design requirements. These documents often contain implicit design assumptions understood only by the person who produced them. Because owners, designers, and contractors interpret documents through different professional perspectives, the same detail can carry different meanings for each stakeholder. Without clear communication, important information is often overlooked or misinterpreted, leading to errors, delays, and added costs. These breakdowns are not solely the result of technical complexity, but also of misaligned incentives and fragmented relationships that discourage transparency and early collaboration (Hoezen et al., 2006). Since project delivery methods determine how

stakeholders interact and how responsibility is assigned, they also influence how openly concerns are communicated. This research paper examines how project delivery methods shape trust, collaboration, and overall stakeholder relationships, and how these social dynamics ultimately influence project outcomes. Understanding this connection is essential for examining how communication functions within each project delivery method and how these structures shape collaboration on construction projects.

This paper argues that project delivery methods do more than set contractual obligations, they also structure the relationships between stakeholders by shaping levels of trust, collaboration, and communication, which directly affect project outcomes. By analyzing these methods through an STS framework, this research demonstrates that delivery methods that better align stakeholder incentives and interactions result in more effective coordination and ultimately improved project outcomes.

BACKGROUND

The main stakeholders in construction projects create a system of risks and responsibilities that interact with one another to progress the project. Owners focus on cost control and project objectives, designers on technical quality, and contractors on constructability, sequencing, and efficient execution. Although these priorities appear aligned, in practice they often create conflict (Yang et al., 2023). This conflict reinforces boundaries between the owners, designers, and general contractors that contribute to miscommunication and information gaps, leading stakeholders to rely on their own interpretations of the same technical documents in an attempt to bridge these gaps. When change orders formal modifications to the original contract scope or cost arise from design misinterpretations, determining responsibility and cost exposure leads to disputes and lengthy negotiations, often resulting in additional delays and costs (Hoezen

et al., 2006). The specifics of the relationships and obligations between stakeholders are determined by the project delivery method, which not only structures how stakeholders communicate and share information, but also shapes the level of trust, collaboration, and alignment between them.

Even though each project is unique, project delivery methods follow standardized industry models that structure stakeholder relationships (Engebø et al., 2020). Design Build, for example, combines the general contractor and design team into one entity, creating a single point of responsibility meaning the owner deals with one party for both design and construction decisions and enabling earlier coordination (Shrestha, 2023). Design Bid Build, on the other hand, separates design and construction, with contractors bidding on a completed design, meaning they submit a price to build something they had no role in designing (Shrestha, 2023). In this structure, the contractor enters with limited insight into design rationale, increasing the likelihood that important assumptions or constraints are not fully communicated. This separation can reinforce fragmented relationships and reduce opportunities for early collaboration (Engebø et al., 2020). Construction Manager at Risk involves the contractor early in design, increasing transparency and facilitating earlier identification of constructability issues meaning problems with how a design can practically be built which can help reduce future design conflicts and change orders (Shrestha, 2023). This earlier involvement can improve communication flow and strengthen alignment between stakeholders. Although CM at Risk and Design Build both involve the contractor early in the design process, the key difference is that the GC is contractually separate from the design team in CM at Risk. This separation makes the owner an intermediary between the designer and the contractor, much like in Design Bid Build. As a result, communication pathways remain partially indirect, which can limit coordination despite earlier

involvement. Recognizing these differences is critical to understanding how each project delivery method shapes communication networks, as well as the underlying relationships that influence collaboration and project performance.

LITERATURE REVIEW

Research on construction project performance shows that delays and cost increases are rarely caused by technical limitations, but instead come from organizational factors that affect coordination between project teams (Liu et al., 2022). A large part of this research focuses on project delivery methods and shows that different methods can lead to different outcomes in terms of cost, schedule, and overall performance (Rigotti et al., 2015).

Across different areas of research, communication is consistently identified as a key factor in project success. Miscommunication is often pointed to as one of the main causes of coordination failures, delays, and inefficiencies (Hoezen et al., 2006). Studies on collaboration and team performance show that when there is limited alignment between team members, transparency and information sharing are reduced, which leads to more problems during construction (Liu et al., 2022). Research on stakeholder relationships also shows that communication improves when roles and responsibilities are clearly defined and when stakeholders feel comfortable raising concerns early in the process (Engebø et al., 2020).

Research on project delivery methods explains how different contractual structures organize how stakeholders interact. Project delivery methods are the structure that organize how different groups interact by contractually defining roles, responsibilities, and communication

between stakeholders. Some methods bring stakeholders together earlier and allow for more coordination, while others keep groups more separated, which can limit early communication and create gaps in understanding (Engebø et al., 2020).

While these areas of research provide useful insights, they are usually studied separately. Studies on project delivery methods tend to focus on performance outcomes, while research on communication, collaboration, and stakeholder relationships focuses on team dynamics (Yang et al., 2023). As a result, there is limited research that directly connects how project delivery methods shape these interactions and how that affects project outcomes (Engebø et al., 2020).

SCOT FRAMEWORK

SCOT (The Social Construction of Technology) is a way of understanding how systems are shaped by the people involved in them. Instead of seeing a system as something fixed or purely technical, this approach focuses on how different groups influence how the system works through their own perspectives and priorities. It looks at how people interpret information and how those interpretations affect how a system functions. This is especially useful in situations where multiple groups are involved and need to work together, since each group may understand the same information in different ways.

In construction projects, the main groups involved are the owner, the designer, and the contractor. Each of these groups has a different role and focus. The owner is concerned with cost, schedule, and overall project goals, the designer focuses on technical quality and design intent, and the contractor is responsible for construction, sequencing, and execution. Even though they are working toward the same project, they approach it from different perspectives and with different priorities. From a SCOT perspective, these groups can be understood as different

groups that interpret the same information in their own way. Because of this, they often approach the same drawings, specifications, or decisions with a different lens, which leads them to interpret the same information differently and can result in confusion, miscommunication, and coordination issues when assumptions are not clearly shared between all parties.

Construction projects are complex systems where a large amount of information is constantly being shared, interpreted, and acted on by different groups. In this type of system, alignment does not happen automatically, and differences in interpretation can lead to ongoing coordination challenges. SCOT helps explain this by showing that these differences are not random, but come from how each group understands the project based on its role. Project delivery methods are the structure that organizes how these groups interact by contractually defining roles, responsibilities, and communication between stakeholders. By shaping when and how stakeholders interact, these structures influence how information is shared and understood, and how easily alignment can be reached. Using this framework, project delivery methods can be analyzed as systems that shape how stakeholders communicate, interpret information, and ultimately work together, which affects overall project outcomes.

ANALYSIS

The data from these studies shows that project delivery methods do affect how projects perform, but not in a way where one method is always the best. Instead, the results vary depending on what is being measured, such as schedule, cost growth, meaning the percentage difference between the original contract amount and the final project cost, administrative effort, and the size of the project. Looking at the data we can see that the chosen delivery method directly impacts project outcomes by changing how teams are organized, how early they interact, and how directly they communicate during the project.

One of the clearest patterns across the data was that Design Build performed better in schedule. In a study of highway projects, schedule growth, meaning the percentage difference between the originally planned project duration and the actual duration, was measured from both the Notice to Proceed date, which is the official date the owner authorizes the contractor to begin work, and the Construction Start date, meaning when physical work on site actually began, which can differ from the authorized start date. Data from over 15,000 projects showed that DB consistently had lower schedule growth than DBB across all cases. On average, DB projects had a mean schedule growth of 12.0% from NTP compared to 17.4% for DBB. From the Construction Start date, DB averaged 13.3% while DBB reached 20.1% (Lampe, 2015). Another study looked at construction intensity, which measures how much work is completed per day relative to the total project scope. DB and CMAR both showed better results than DBB, meaning that work was completed at a faster rate once construction began (Shrestha, 2023). Taken together, the schedule growth and construction intensity data showed that delivery methods with earlier contractor involvement or a closer connection between design and construction allowed work to move more efficiently by reducing delays related to constructability, sequencing, meaning the order in which construction tasks are planned and executed, and material availability. Because these questions could be handled earlier, they were less likely to interrupt work in the field.

Having early involvement is not always necessary and can sometimes be excessive. In smaller projects, DBB sometimes performs slightly better than DB and CMAR. For projects under \$2 million, DBB had an average cost growth of 0.58% compared to 0.87% for DB. However, this changes as the project size increases. In projects between \$10 million and \$20 million, DB had a mean cost growth of 2.58% compared to 3.77% for DBB. In projects between

\$20 million and \$50 million, DB averaged 4.33% while DBB reached 5.87%. For projects larger than \$50 million, DB continued to perform better with 5.93% compared to 6.94% for DBB (Lampe, 2015). This pattern shows that there is no single project delivery method that is always better. The effectiveness of each method depends on how much coordination the project requires between the owner, designer, and contractor. In smaller projects, the scope and design are usually easier to define, so DBB can still work well because the separation between design and construction creates less coordination risk. As projects become larger, that separation becomes more difficult to manage because cost, sequencing, constructability, and design decisions affect more parts of the project. This explains why DB becomes more effective in larger projects, where the closer connection between design and construction helps reduce coordination gaps before they become more serious during construction.

The data also points to a broader pattern beyond cost and schedule alone. Across the studies, certain factors appear consistently in projects that perform better: clear communication, trust between stakeholders, early involvement, and strong coordination between teams (Shrestha, 2023). These factors appear more often in DB and CMAR projects and less so in DBB, where the structure keeps teams more separated. This pattern reflects how each delivery method defines the boundaries between teams, how freely information moves, and how quickly problems get addressed. Methods that reduce that separation consistently show better schedule performance and more stable cost outcomes, and those differences become more pronounced as project complexity increases. Ultimately, this shows that delivery method performance is not just a function of contract structure, but also depends on the conditions each method creates for communication, coordination, and trust to develop between the people doing the work.

The differences in performance between project delivery methods are connected to how each method structures interaction between stakeholders. SCOT argues that systems are not shaped by technical logic alone, but by the social groups that interact with them and the meanings those groups bring to them (Bijker, 2001). In construction projects, the owner, designer, and contractor act as different relevant social groups, each approaching the project with different priorities, assumptions, and definitions of success. This is what SCOT calls interpretive flexibility, where the same project can mean something different depending on which group is interpreting it. The data shows that when these groups are more separated, as in Design Bid Build, they are more likely to move forward with different understandings of the project because the sequential structure gives them fewer opportunities to resolve those differences early. These unresolved differences then show up in the results as higher schedule growth, more delays, and the need for adjustments during construction.

Methods like Design Build and CMAR performed better because they changed when and how stakeholders were able to reach a shared understanding of the project. In SCOT, this process is called closure, meaning that different interpretations become more stable and aligned. Design Build removes more separation by combining design and construction, while CMAR reduces some of that separation by bringing the contractor in earlier. Both structures allow stakeholders to interact earlier and more directly than in DBB, creating conditions where assumptions can be clarified, concerns can be raised, and differences can be resolved before decisions become costly to change. This earlier alignment helps explain why these methods show better schedule performance and more stable cost outcomes, especially as projects grow larger and require more coordination.

The data shows that delivery methods affect performance, but SCOT explains why. Each method does not only organize contracts differently. It also determines when and how the relevant social groups resolve differences, negotiate responsibilities, and develop a shared understanding of the project. The choice of delivery method is not only a procedural or contractual decision, but also a decision about how differences between stakeholders will be managed and when those differences will be addressed during the project.

COUNTERARGUMENTS

One counterargument is that methods with fewer boundaries are not automatically better because they can create new risks when responsibility becomes unclear. In Design Build, the closer connection between the designer and contractor can improve coordination, but it can also make it harder to determine who is responsible when a problem arises. If there is a design problem, construction issue, cost increase, or unclear scope, meaning an incomplete or disputed definition of what work is included in the contract, stakeholders may disagree over whether it came from design decisions, construction execution, or owner expectations. This can create conflict, particularly because risk management, meaning how responsibility for unexpected costs or problems is assigned between the owner, designer, and contractor, becomes more complicated when roles overlap. DB and CMAR projects can face issues such as owner lack of control over design, lack of owner knowledge about the delivery method, design problems, personnel issues, and costly changes (Shrestha, 2023). This does not disprove the argument that reduced boundaries improve communication, but it shows that those benefits depend on clear expectations, strong owner involvement, and well-defined responsibility.

Another counterargument is that project outcomes may depend as much on stakeholder experience as on the delivery method itself. A delivery method can create the structure for

communication, but it does not guarantee that the owner, designer, and contractor will use that structure effectively. If the owner does not understand how DB or CMAR changes their role, or if the project team lacks experience with early collaboration, the method may fail to produce the trust and coordination it is meant to support. Owner education, staff training, leadership, teamwork, and communication were key factors in making DB and CMAR projects successful (Shrestha, 2023). This means that delivery methods matter because they shape the opportunity for collaboration, but stakeholder experience determines whether that opportunity turns into better communication, trust, and project performance.

CONCLUSION

Looking at all of the data and analysis together, it becomes clear that project delivery methods do influence how projects perform, but not in a simple way where one method is always better than the others. Instead, the results show consistent patterns, where Design Build performs best in schedule, CMAR shows strong cost control and stability, and Design Bid Build tends to perform worse as projects become more complex. At the same time, the data also shows that these differences are not only tied to the delivery method itself, but to how each method structures the way stakeholders interact throughout the project.

Methods like Design Build and CMAR reduce the boundaries between the owner, designer, and contractor, allowing for earlier involvement and more direct communication. This leads to better alignment, fewer delays, and more controlled outcomes. In contrast, Design Bid Build creates more separation between these groups, which increases the likelihood of miscommunication, delays, and the need for adjustments later in the project. While DBB can perform similarly in smaller projects, where coordination is more limited, the data shows that it becomes less effective as the level of complexity increases.

Applying SCOT, these results show that differences in performance come from how each method shapes the way stakeholders interpret and interact with the project. When stakeholders are more integrated, they are able to develop a shared understanding earlier, which reduces misalignment and improves coordination. As a result, project delivery methods that encourage stronger interaction and reduce separation between teams create conditions where trust and alignment can develop, leading to better overall outcomes. This shows that improving project performance is not only about selecting a delivery method, but about understanding how that method influences the relationships between the people involved in the project.

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